Introduction

COMPSCI 485, Applications of Natural Language Processing

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What this course is about

- Fundamental principles in natural language processing, and how to do natural language processing in practice
  - Hands-on: data collection and implementation
  - Discussion of basic linguistic issues
  - Know when NLP works and when it doesn’t
- Course topics include: text processing, corpus data, probabilistic language models, syntax models, word embeddings, text classification, sentiment analysis, question answering, deep learning methods, large language models...

Natural language processing (NLP) in practice

- What natural language processing (NLP) applications do you know?
- Have you used them yourself?
- What would you say, how well do they work?

- First discuss in small groups, 4 to 5-ish
- Then we'll talk in the whole class

Some NLP tasks on my list

- Spell checking, grammar checking
- Speech interfaces
- Dialogue agents / voice assistants
- Search and summarization
- Machine translation
- Sentiment analysis

Ambiguity in language

One reason why NLP is hard

Camilla's Team Rubbish Claims She is the Royal Racist

Hospitals named after sandwiches kill five

Language Log,
https://languagelog.ldc.upenn.edu

Levels of linguistic structure

This is a simple sentence.

Levels of linguistic structure

Characters

T	h	i	s		i	s		a		s	i	m	p	l	e		s	e	n	t	e	n	c	e	.

Levels of linguistic structure

Morphology

This	is	a	simple	sentence	.
	be **3sg** **present**				

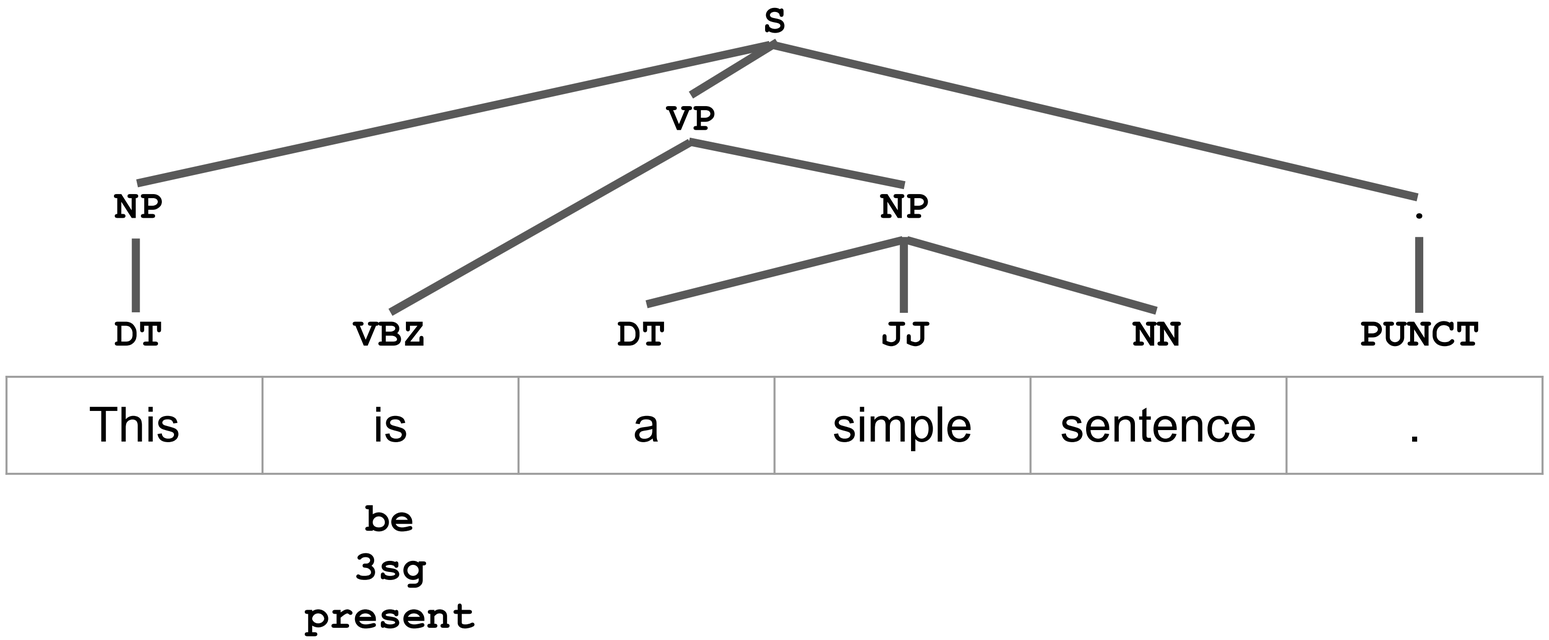
Levels of linguistic structure

Parts of speech

DT	VBZ	DT	JJ	NN	PUNCT
This	is	a	simple	sentence	.
	be 3sg present				

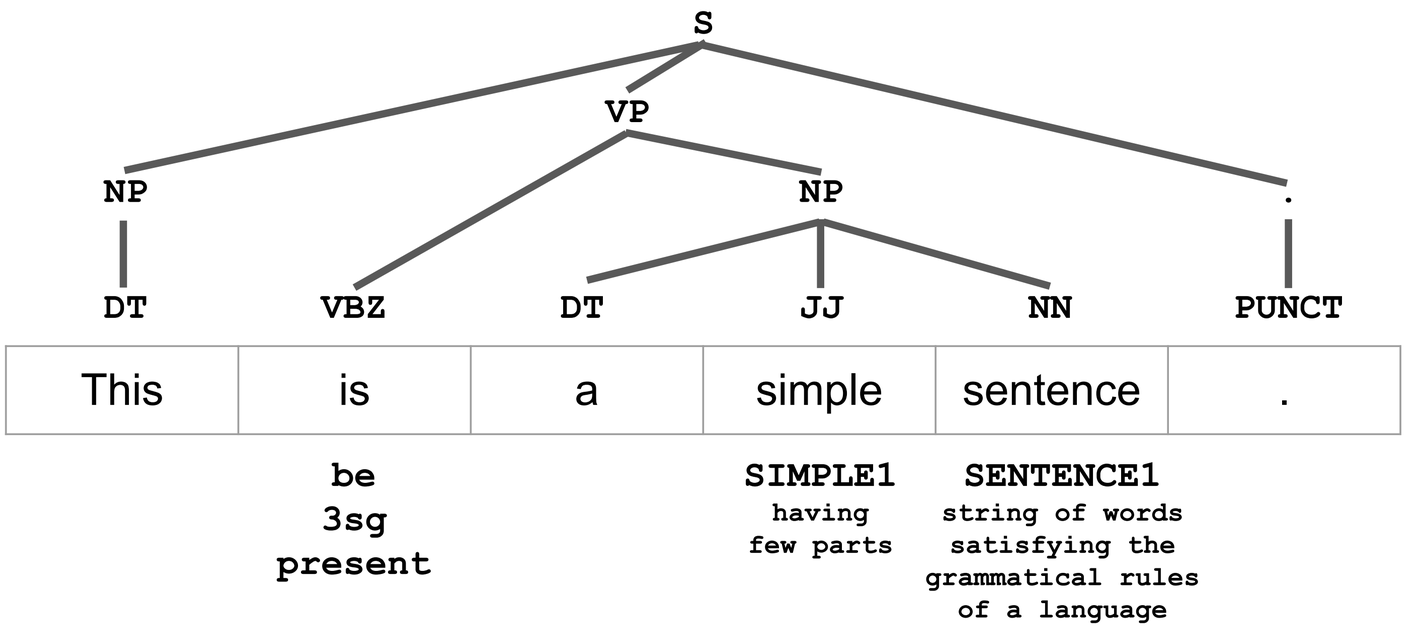
Levels of linguistic structure

Syntax



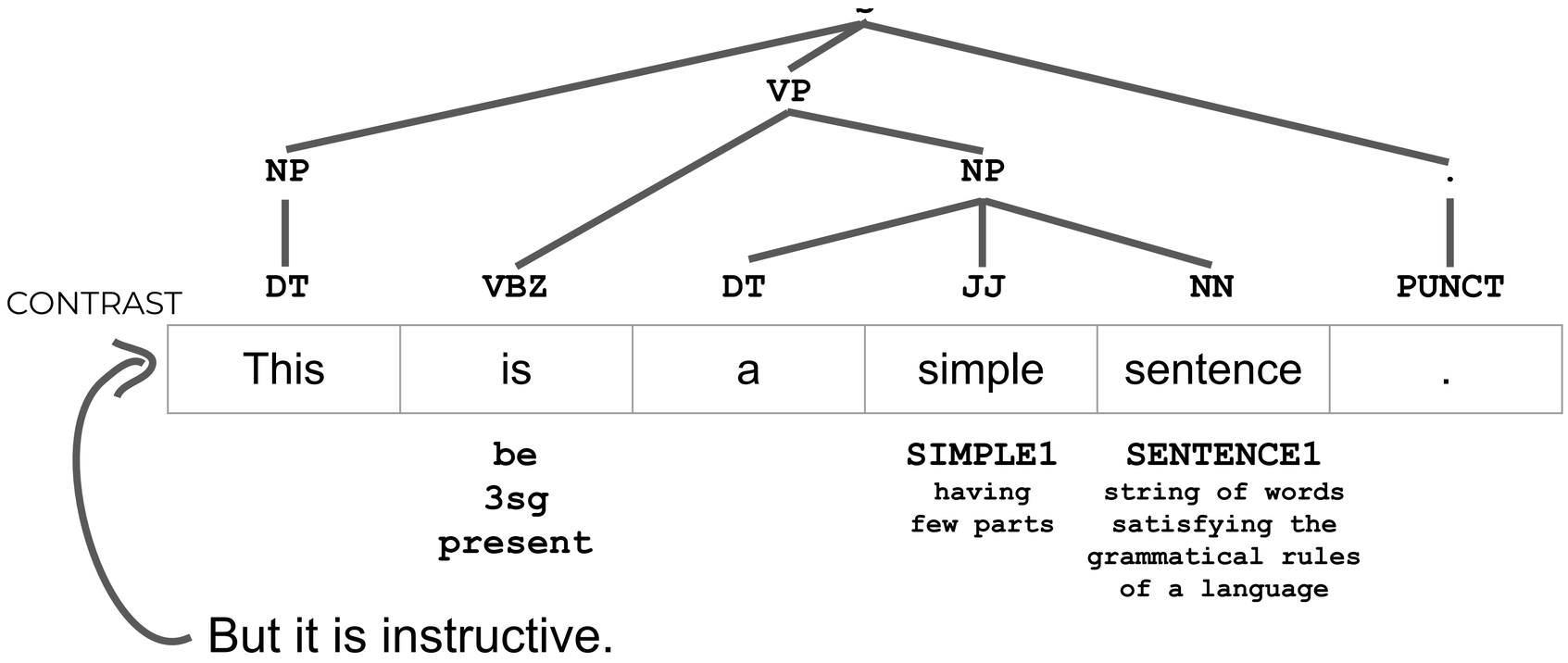
Levels of linguistic structure

Semantics



Levels of linguistic structure

Discourse



Back to our earlier cases of ambiguity

- **The first is about part of speech:**
  Camilla's Team Rubbish Claims She is the Royal Racist
  Verb?Noun? Verb?Noun?

- **The second is about syntax:**
  [Hospitals named] after [sandwiches kill five]
  [Hospitals named after sandwiches] kill five

- **The third is about syntax and word meaning:**
  Driving [waste and recycling] innovation
  [Driving waste] and [recycling innovation]
  Driving: leading? giving a ride?

Large language models, e.g. ChatGPT

MA

Rewrite the following paragraph in the style of a breezy, readable airport book:

A spectre is haunting Europe — the spectre of communism. All the powers of old Europe have entered into a holy alliance to exorcise this spectre: Pope and Tsar, Metternich and Guizot, French Radicals and German police-spies.

Picture this: Europe, a continent buzz
shadows is a haunting spectre, none o
that sends shivers down the spines of
forming an unlikely holy alliance to ba
everyone on board: the Pope and the Tsar, Metternich and Guizot, even the French Radicals and those sneaky German police-spies. It's a real clash of the titans, and they're ready to wage a battle against this spooky ideology. Hold on tight, because things are about to get interesting!

From other samples

- [...] with some seriously big names. We're talking the Pope and the Tsar, Metternich and Guizot, and even those fiery French Radicals and crafty German police-spies.
- Picture this: the Pope and the Tsar, Metternich and Guizot, even those French Radicals and German police-spies joined forces!

https://twitter.com/MattCowgill/status/1669506317961273344

Where is NLP at right now?

- What is possible now that wasn’t possible a short time ago?
- What is still difficult?

Course info: Requirements

- (10%) Quizzes / exercises (approx. weekly)
- (30%) Problem sets
  - Written: math and concepts
  - Programs: in Python
- (30%) In-class Midterm
- (30%) Final projects (groups of 2-3) [Choose a topic, or select a suggested topic]
  - Project Proposal
  - Progress Report
  - In-class presentations
  - Final Report

Technologies

- Canvas page has schedule and readings
- Assignments will be on Gradescope
- Discussions: Piazza
- Echo360 for lecture capture, available through Canvas

Upcoming assignments

- First “exercise” will be made available to day through Gradescope: a survey

- First homework will be made available next week, due Feb 24

Approximate list of class topics in order

- Introduction
- Words and regular expressions
- Text Classification and Naive Bayes
- Logistic Regression Classification
- Evaluating classifier performance
- Annotations
- Text Data, Final Project Overview
- Tagging
- Context-Free Grammars
- Parsing
- Midterm review
- Midterm in class
- Dependency parsing
- n-Gram Language Models
- Word Embeddings
- Neural networks
- Transformers and BERT
- Generative LLMs (incl. GPT)
- Inference and question answering
- Retrieval-augmented generation
- Final presentations

Tasks coming up

- Exercise on gradescope, homework